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```
getwd()
```

```
[1] "C:/Users/sims student 18/Desktop"
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```
disas<- read.csv("dataset.csv")
disas
```

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```
head(disas)
```

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```
tail(disas)
```

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```
str(disas)
```

```
'data.frame':  20000 obs. of  13 variables:
 $ event_id      : int  1 2 3 4 5 6 7 8 9 10 ...
 $ disaster_type : chr  "Wildfire" "Hurricane" "Volcanic Eruption" "Drought" ...
 $ location      : chr  "Chile" "India" "Italy" "Chile" ...
 $ latitude      : num  -34.7 22.1 42.3 -33.4 39.4 ...
 $ longitude     : num  -71.8 78 11 -70 37 ...
 $ date          : chr  "27-08-2025" "29-05-2023" "15-01-2023" "08-02-2024" ...
 $ severity_level: int   8 5 7 8 8 9 7 7 4 4 ...
 $ affected_population : int 31104 29340 34804 31191 46284 49656 33514 34713 8039 20776 ...
 $ estimated_economic_loss_usd: num 2768213 5996227 9222541 1827703 13435921 ...
 $ response_time_hours : num  5.12 44.43 49.3 65.56 60.96 ...
 $ aid_provided      : chr  "Yes" "No" "No" "Yes" ...
 $ infrastructure_damage_index: num  0.59 0.26 0.94 0.94 0.92 0.95 0.4 1 0.53 0.64 ...
 $ is_major_disaster  : int   1 0 1 1 1 1 1 1 0 0 ...
```

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```
summary(disas)
```

event_id	disaster_type	location	latitude
Min. : 1	Length:20000	Length:20000	Min. : -43.13
1st Qu.: 5001	Class :character	Class :character	1st Qu.: 3.56
Median :10000	Mode :character	Mode :character	Median : 24.36
Mean :10000			Mean : 18.75
3rd Qu.:15000			3rd Qu.: 38.29
Max. :20000			Max. : 48.45
longitude	date	severity_level	affected_population
Min. : -102.134	Length:20000	Min. : 1.000	Min. : 0
1st Qu.: 8.347	Class :character	1st Qu.: 3.000	1st Qu.:14755
Median : 74.600	Mode :character	Median : 5.000	Median :27613
Mean : 41.977		Mean : 5.489	Mean :27641
3rd Qu.: 117.461		3rd Qu.: 8.000	3rd Qu.:40017
Max. : 145.903		Max. :10.000	Max. :75147
estimated_economic_loss_usd	response_time_hours	aid_provided	
Min. : 0	Min. : 1.00	Length:20000	
1st Qu.: 2024090	1st Qu.:18.41	Class :character	
Median : 4031418	Median :36.56	Mode :character	
Mean : 4831073	Mean :36.37		
3rd Qu.: 7018268	3rd Qu.:54.02		
Max. :21868935	Max. :71.99		
infrastructure_damage_index	is_major_disaster		
Min. :0.0600	Min. :0.0000		
1st Qu.:0.4000	1st Qu.:0.0000		
Median :0.5500	Median :0.0000		
Mean :0.5573	Mean :0.4001		
3rd Qu.:0.7000	3rd Qu.:1.0000		
Max. :1.0000	Max. :1.0000		

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```
ls(disas)
```

```
[1] "affected_population"      "aid_provided"
[3] "date"                    "disaster_type"
[5] "estimated_economic_loss_usd" "event_id"
[7] "infrastructure_damage_index" "is_major_disaster"
[9] "latitude"                 "location"
[11] "longitude"                "response_time_hours"
[13] "severity_level"
```

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```
disas$disaster_type<-as.factor(disas$disaster_type)
```

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```
disas$location<-as.factor(disas$location)
```

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```
disas$date<-as.factor(disas$date)
```

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```
disas$aid_provided<-as.factor(disas$aid_provided)
```

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```
colSums(is.na(disas))
```

```
      event_id      disaster_type
      0          0
location      latitude
      0          0
longitude      date
      0          0
severity_level affected_population
      0          0
estimated_economic_loss_usd response_time_hours
      0          0
aid_provided infrastructure_damage_index
      0          0
is_major_disaster
      0
```

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```
sum(disas$estimated_economic_loss_usd)
```

```
[1] 96621452916
```

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```
mean(disas$estimated_economic_loss_usd)
```

```
[1] 4831073
```

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```
aggregate(infrastructure_damage_index~affected_population ,data=disas,sum)
```

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```
aggregate(infrastructure_damage_index~is_major_disaster ,data=disas,sum)
```

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```
min(disas$affected_population)
```

```
[1] 0
```

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```
max(disas$affected_population)
```

```
[1] 75147
```

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```
subset.data<-subset(disas,select=c("latitude","longitude","severity_level","response_time_hours"))
str(subset.data)
```

```
'data.frame':  20000 obs. of  4 variables:
 $ latitude      : num  -34.7 22.1 42.3 -33.4 39.4 ...
 $ longitude     : num  -71.8 78 11 -70 37 ...
 $ severity_level : int   8 5 7 8 8 9 7 7 4 4 ...
 $ response_time_hours: num   5.12 44.43 49.3 65.56 60.96 ...
```

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```
mean(subset.data$latitude)
```

```
[1] 18.74981
```

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```
median(subset.data$response_time_hours)
```

```
[1] 36.56
```

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```
mode(subset.data$severity_level)
```

```
[1] "numeric"
```

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```
sd(subset.data$latitude)
```

```
[1] 25.02676
```

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```
sd(subset.data$longitude)
```

```
[1] 82.53592
```

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```
sd(subset.data$severity_level)
```

```
[1] 2.866269
```

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```
sd(subset.data$response_time_hours)
```

```
[1] 20.42057
```

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```
max(subset.data$longitude)
```

```
[1] 145.9027
```

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```
min(subset.data$longitude)
```

```
[1] -102.1337
```

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```
quantile(subset.data$severity_level,prob=c(.20,.50,.45,.95))
```

20%	50%	45%	95%
2	5	5	10

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```
quantile(subset.data$longitude,prob=c(.20,.50,.45,.95))
```

20%	50%	45%	95%
-70.94747	74.60032	35.89825	138.67726

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```
quantile(subset.data$latitude,prob=c(.20,.50,.45,.95))
```

20%	50%	45%	95%
-0.3423391	24.3610062	20.9863105	42.5542827

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```
quantile(subset.data$response_time_hours,prob=c(.20,.50,.45,.95))
```

20%	50%	45%	95%
15.0300	36.5600	32.9855	68.1605